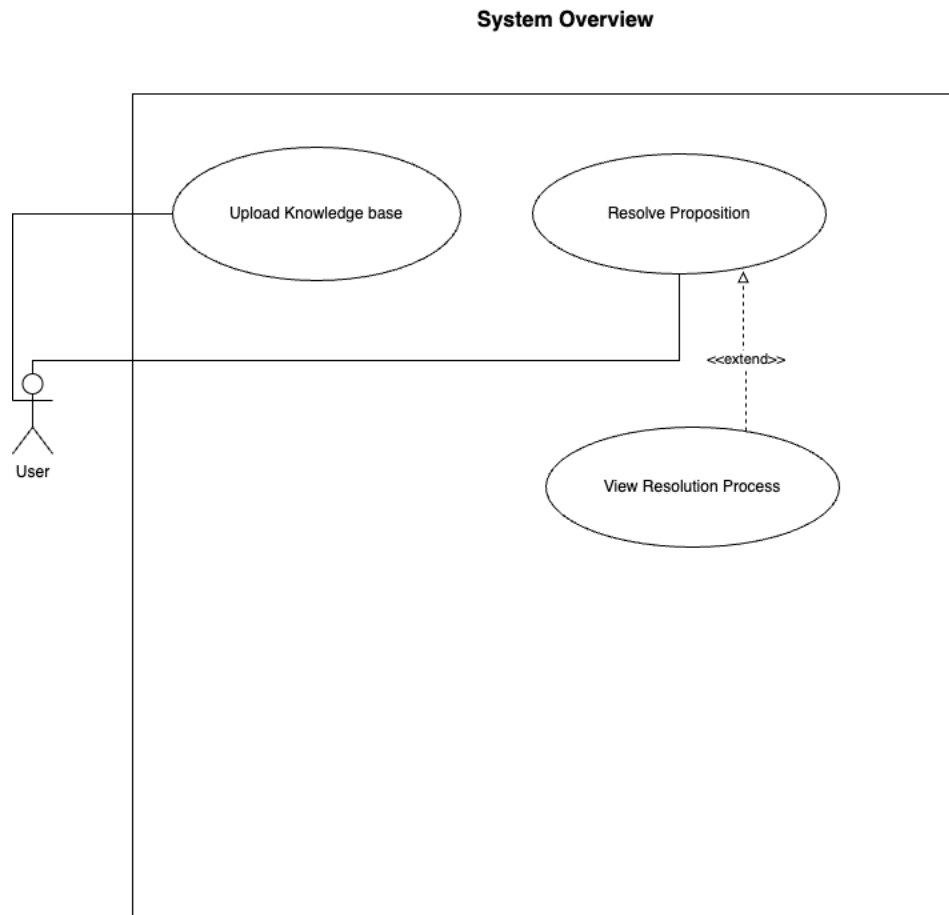


Use Case Narratives:

System overview:

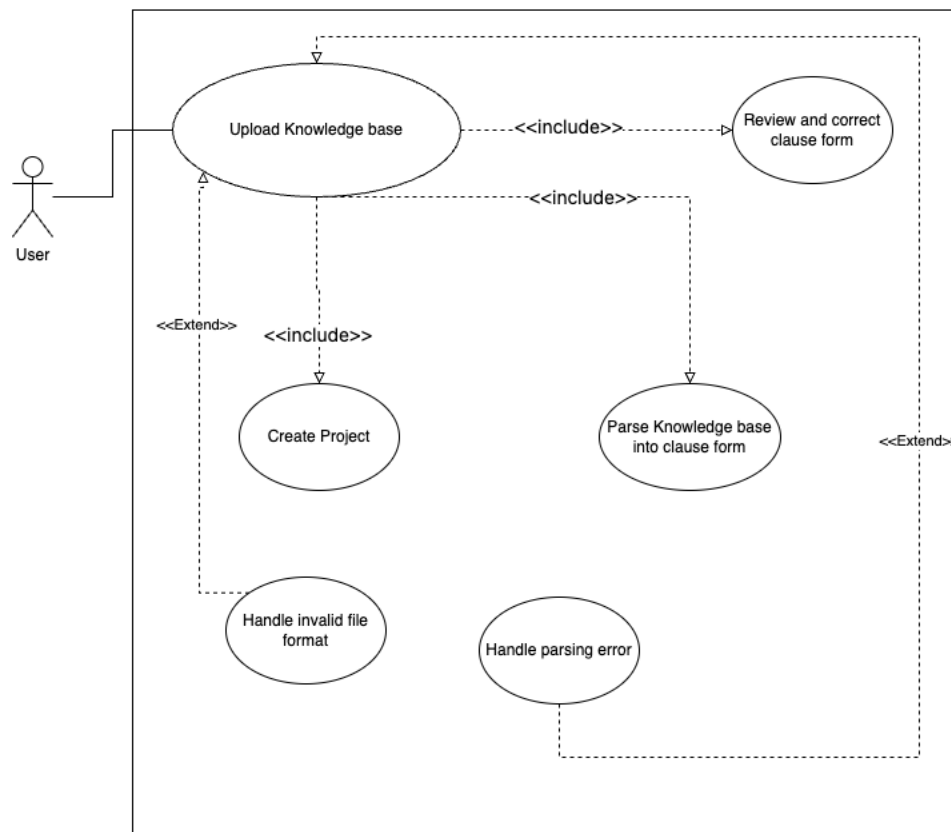


Upload a knowledge base — Primary actor: User

User creates a new project and names it. User selects a knowledge base file and uploads it. System parses the file into clause form. User reviews the resulting clauses and corrects any errors. User confirms the knowledge base, and System activates it for the project.

If the uploaded file is not in the required format, System rejects it and asks User to select a valid file. If System cannot parse the file into valid clause form, System reports the error and indicates the affected section so User can correct the source file and re-upload. User may leave the review stage without confirming, in which case the knowledge base stays inactive.

Upload a knowledge base Use Case Diagram



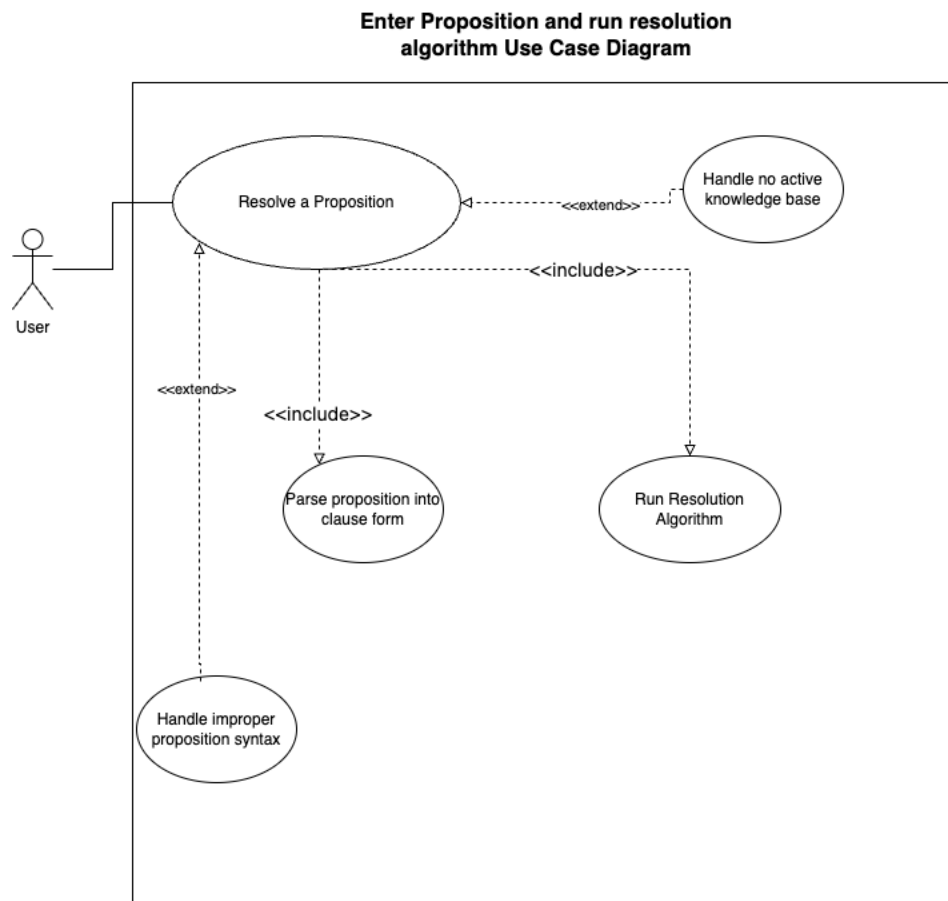
Enter Proposition and run resolution algorithm– Primary actor: User

Precondition: An active knowledge base exists for the project

User enters a proposition to test. System parses the proposition into clause form. System runs the resolution algorithm against the active knowledge base. System determines the conclusion and displays whether the proposition is proven, refuted, or indeterminate. User reviews the conclusion.

If the proposition entered is not in the required format, the system rejects it and asks the user to enter a valid clause.

If no knowledge base is active, the system asks the User to upload and activate one before a proposition can be entered. If the proposition is not in valid syntax, the system reports the parsing error and asks the user to correct it. If resolution cannot reach a definitive result within its limits, System reports the conclusion as indeterminate rather than proven or refuted.



View resolution process — Primary actor: User. Extends: *Resolve a proposition*.

User clicks view resolution steps. System displays the resolution steps as a list of clause comparisons, highlighting the one currently being viewed. The User then steps forward and backward through the comparisons. When a comparison proves the proposition impossible, System highlights it in red, then displays the conclusion — red if impossible, green if possible. User returns to the previous screen.

At any point, User may switch to a tree view of the same comparisons, and steps forward and backward within it the same way. User returns to the previous screen from either view.

View Resolution

